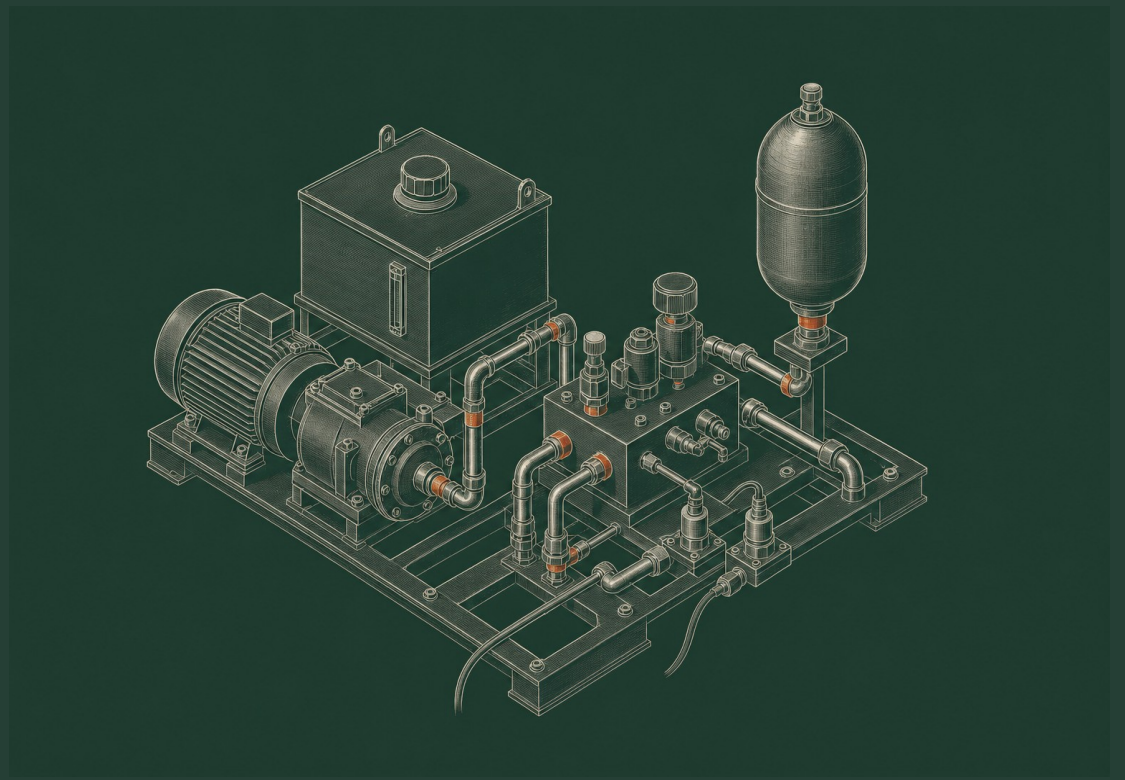
Detecting Hydraulic Pump Leakage from Sensor Signals

Comparing RAW, FIR, and IIR signals with feature-based machine learning



Concept illustration, not a photograph or model of the actual test rig.

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Dataset: UCI Condition Monitoring of Hydraulic Systems [1]

Pressure and flow measurements need context

Sensors record how the rig operates. Fault labels come from the dataset and guide model training.

DE / DATA INTEGRITY

Keep sensor values and timing consistent

60 seconds

PS1 and FS1 use different sampling rates and units.

DSP / REPRESENTATION

Compare features from raw and filtered signals

14 features

Filters change fluctuations and transients. We test whether those changes help.

DSA / VALIDATION

Measure predictions and error types

3 classes

Normal, mild, severe. Error types matter.

A smoother trace may hide a useful fault signal.
We test filtering against the original recordings.

Source: UCI data [1]; preprocessing and evaluation notebooks for this study.

Does filtering help classify pump leakage?

Can FIR or IIR produce more useful classification features than the unfiltered signals?

Smoothing can reduce fluctuations, but it can also blur rapid changes that matter.

RAW	FIR	IIR
Unfiltered baseline	Causal moving average	Exponential smoothing

Same 14 features, same Random Forest settings,
and the same grouped validation splits.

Measures: mean Macro F1, fold variation, and missed severe-leakage cases

Source: experiment design and evaluation notebooks for this study.

How the sensor recordings become model inputs

Each observation keeps its sensor, sampling rate, cycle ID, and leakage label.

Stage	What I used or checked	Why it matters	Evidence in this study
Recorded signals	Digital pressure and flow measurements	Two sensors run at different sample rates	PS1: 100 Hz FS1: 10 Hz
Quality checks	Check NaN, infinity, and zero readings	Avoid treating valid zeros as missing data	No NaN or infinity in PS1 or FS1
Cycle matching	Keep each sensor tied to its 60-second cycle	Sample index is not the same as time	Join by cycle ID, not raw sample index
Feature table	Calculate seven features per sensor and cycle	Each cycle becomes one model observation	1,449 cycles with 14 input features

All processing uses existing digital recordings.
Live sensing and deployment were not tested.

Sources: UCI hydraulic dataset [1]; preprocessing notebooks for this study.

Two sensors observe the same cycle at different rates

PS1 and FS1 cover the same 60 seconds but take readings at different intervals.

PROBLEM / DIFFERENT RATES

PS1: 6,000 readings
FS1: 600 readings

60 seconds

The same sample index does not mean the same time.

→

METHOD / KEEP TIME

Use each sensor's own time axis

1 cycle

Check data → filter by cycle → calculate sensor features.

→

OUTPUT / JOIN BY CYCLE

Combine seven PS1 and seven FS1 features

1 row

One feature row for each 60-second cycle.

At index 100, PS1 is at 1 second while FS1 is at 10 seconds. Matching raw array indices would mix different times.

Source: sampling rates in UCI [1]; data preparation notebooks for this study.

What the 14 features capture

For each sensor, five features describe time behavior and two describe frequency content.

Signal property	Features used	What it describes
Typical level	Mean, RMS	Average and effective signal magnitude
Variability	Standard deviation, peak-to-peak	How widely the values change within a cycle
Fast changes	Difference RMS	Size of changes between adjacent samples
Frequency content	Spectral centroid, relative spectral energy	How energy is distributed over frequency

Seven PS1 features plus seven FS1 features give 14 inputs.
Random Forest stays fixed across RAW, FIR, and IIR.

Sources: feature extraction notebooks; SciPy FFT and signal processing [3].

An overall score can hide missed severe faults

A high Macro F1 score can coexist with severe leakage being classified as mild.

Features	Prediction	Error review	Reporting	Decision support
14 values per cycle	Normal, mild, or severe	Which classes get confused?	Explain error counts and limits	Inspect findings before acting

Check: class recall, class precision, and the out-of-fold confusion matrix

Missed severe leakage and incorrect severe predictions create different concerns.

For engineers

Check sensor integrity, filter response, and errors.

For managers

Use results to plan further validation, not automatic action.

Source: evaluation notebooks; precision, recall, and grouped validation [4].

One 60-second cycle is one observation

UCI Hydraulic Systems contains recordings from one laboratory test rig.

2,205

cycles in the full dataset
not individual time samples

Sensor	Measurement and rate	Samples per cycle
PS1	Pressure (bar) 100 Hz	6,000 samples 60 seconds
FS1	Flow (L/min) 10 Hz	600 samples 60 seconds

1,449

stable cycles in this analysis
with nearly balanced classes

0 · Normal	1 · Mild	2 · Severe
489	**480**	**480**

Target: pump leakage class 0 / 1 / 2.
UCI supplies digital data. We did not run an ADC.

Source: UCI Hydraulic Systems [1]; Helwig et al. (2015) [2].

Checking the recorded pressure and flow traces

Recorded PS1 and FS1 signals across three leakage classes, with other component settings matched.

01. Check data quality

No NaN or infinity in PS1 or FS1.
Zero flow is not automatically missing.

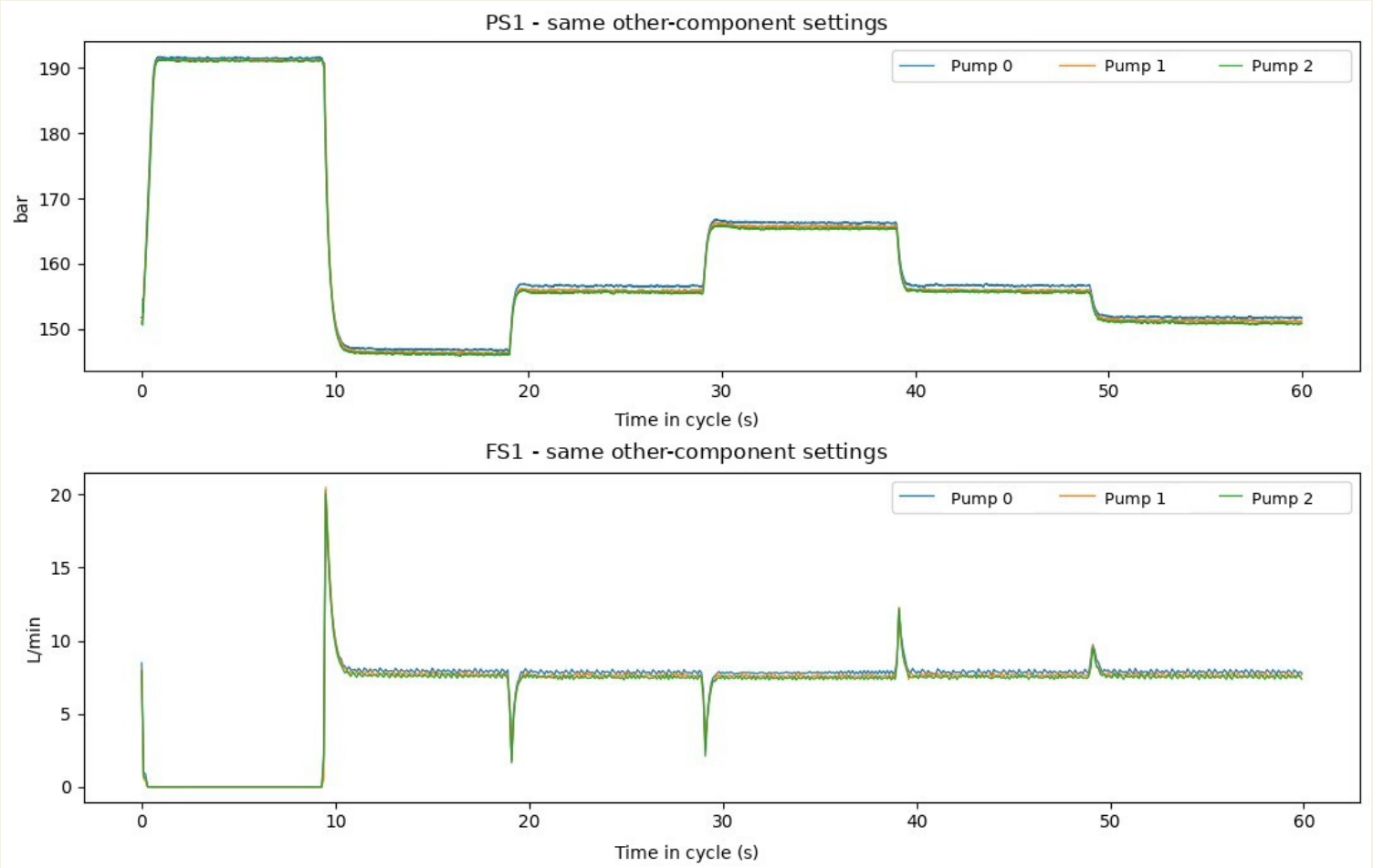
02. Respect sampling rates

PS1: Δt = 10 ms; Nyquist 50 Hz.
FS1: Δt = 100 ms; Nyquist 5 Hz.

Align by time and cycle,
not by sample index.

03. Build the feature table

Filter each cycle → summarize both sensors → one feature row.



Sources: waveform plots and audit notebooks; UCI sampling rates [1].

FIR and IIR change the shape of rapid transitions

Recorded PS1 and FS1 responses near a large change in cycle #2190.

RAW: baseline

Keeps the measured fluctuations and transients without smoothing.

FIR: no feedback

$y[n] = (1/L)\ \Sigma\ x[n-k]$

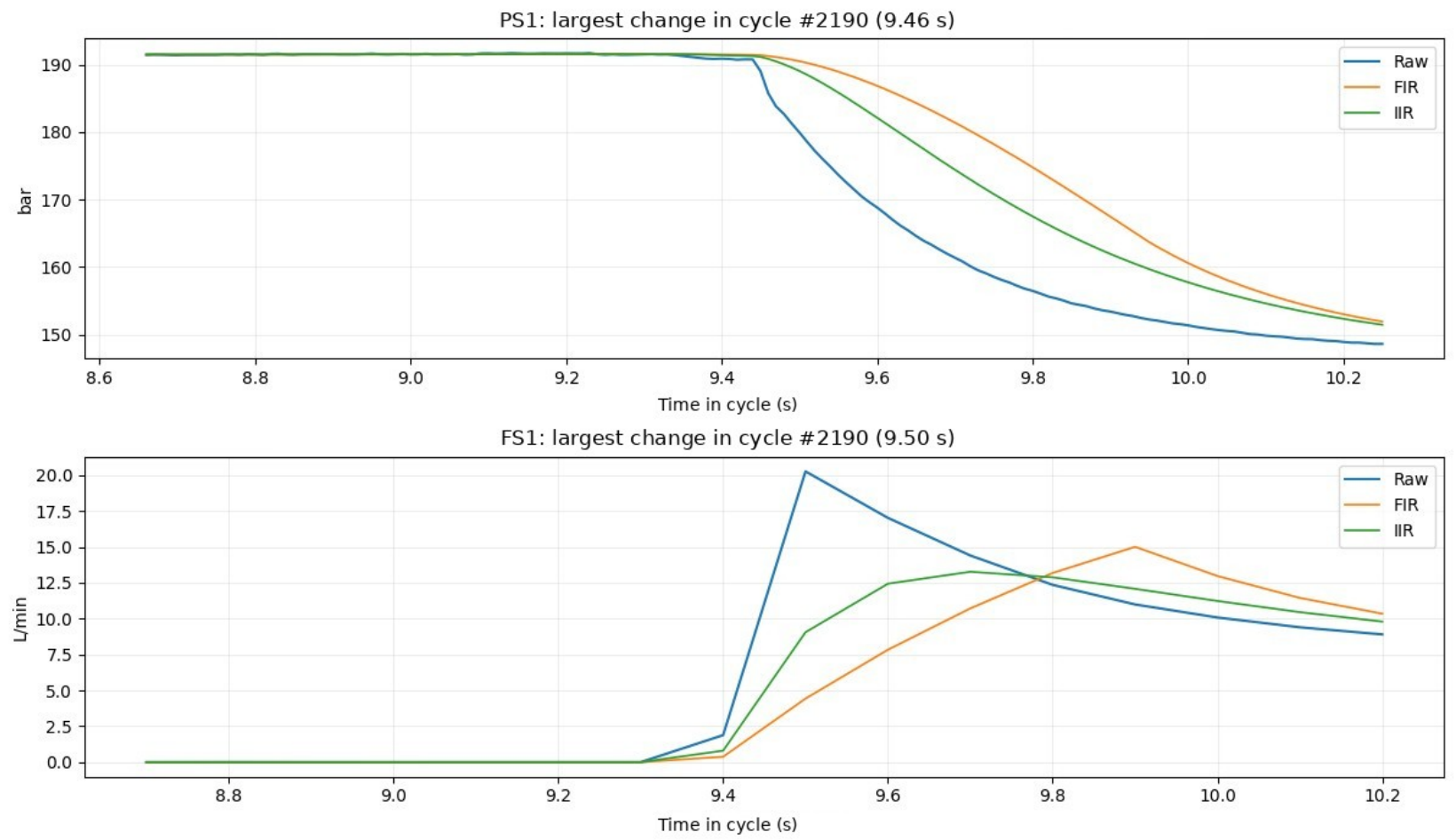
$k = 0, \ldots, L-1$

Average of the current and previous $L-1$ samples.

IIR: feedback

$y[n] = (1-a)x[n] + a \cdot y[n-1]$

Uses the previous output. Changes decay gradually.



Sources: filter-comparison notebook; SciPy signal filtering [3].

H(z) describes the digital filter response

LTI: $y[n] = x[n] * h[n]$ | $H(z) = Y(z)/X(z)$ | z^{-1} delays by one sample.

FIR: finite impulse response

$H(z) = (1/L)\ \Sigma[k=0\ldots L-1]\ z^{-k}$

Poles at 0; zeros where $z^L=1$, except $z=1$.

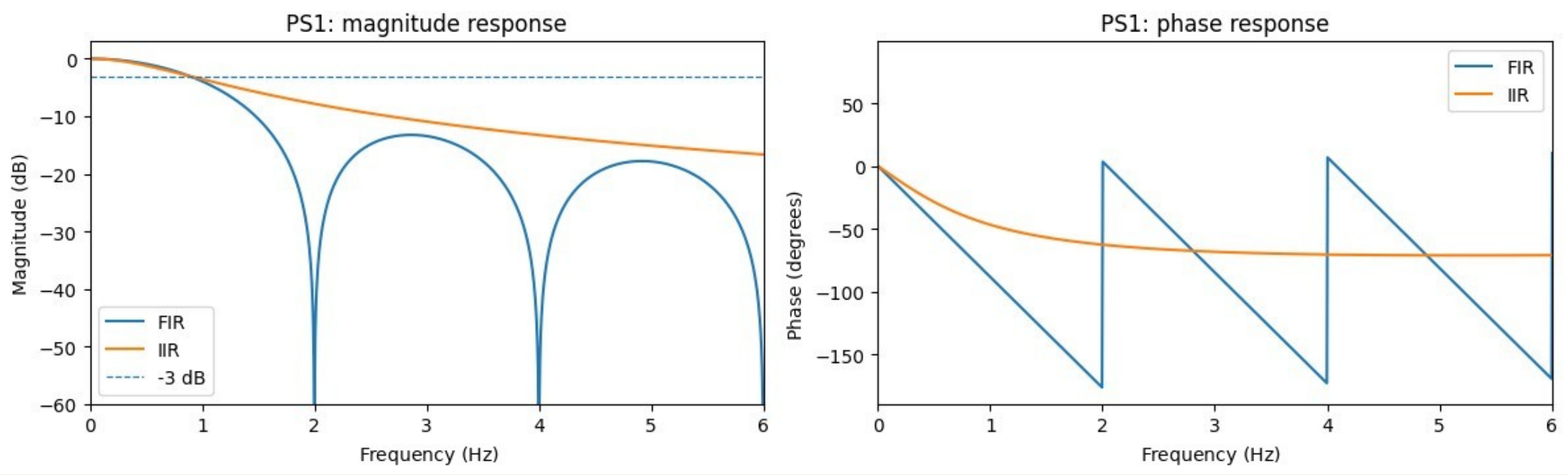
Finite coefficients give a stable filter.

IIR: decaying impulse response

$H(z) = (1-a) / (1-az^{-1})$

Pole at $z=a$; zero at $z=0$.

$0 < a < 1$ places the pole inside the unit circle.



Sources: filter frequency-response notebook; SciPy freqz [3].

Group-aware validation keeps configurations separate

One cycle gives 14 features. Each signal version uses the same Random Forest classifier.

Feature engineering

Domain	Features per sensor
Time (5)	mean, SD, RMS, peak-to-peak, diff RMS
Frequency (2)	spectral centroid, relative spectral energy

RAW / FIR / IIR → 14 features
→ Random Forest → held-out predictions

5-fold GroupKFold

	G1	G2	G3	G4	G5
Fold 1	**Test**	Train	Train	Train	Train
Fold 2	Train	**Test**	Train	Train	Train
Fold 3	Train	Train	**Test**	Train	Train
Fold 4	Train	Train	Train	**Test**	Train
Fold 5	Train	Train	Train	Train	**Test**

Groups are defined by cooler, valve, and accumulator states.
G1-G5 show the split idea, not the actual number of groups.
This does not test generalization to other machines.

Metrics: Macro F1 / severe recall / OOF confusion matrix / fold-to-fold variation

Source: experiment notebooks; scikit-learn GroupKFold [4].

IIR has the highest mean Macro F1, by 0.83 points

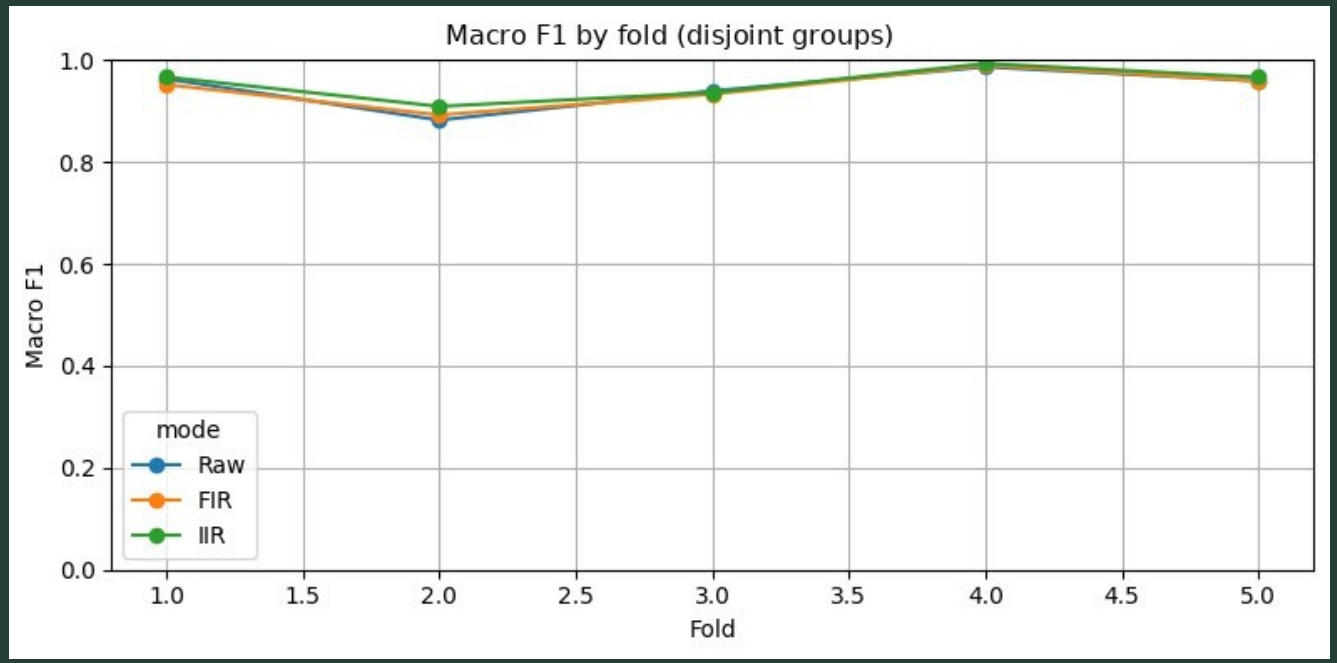
Five grouped folds. SD measures variation across folds, not uncertainty in the population.

Method	Mean F1	SD
RAW	**94.62%**	3.94 pp
FIR	**94.54%**	3.60 pp
IIR	**95.45%**	3.23 pp

+0.83 pp

IIR vs. RAW. FIR is 0.08 percentage points below RAW.

Macro F1 by fold



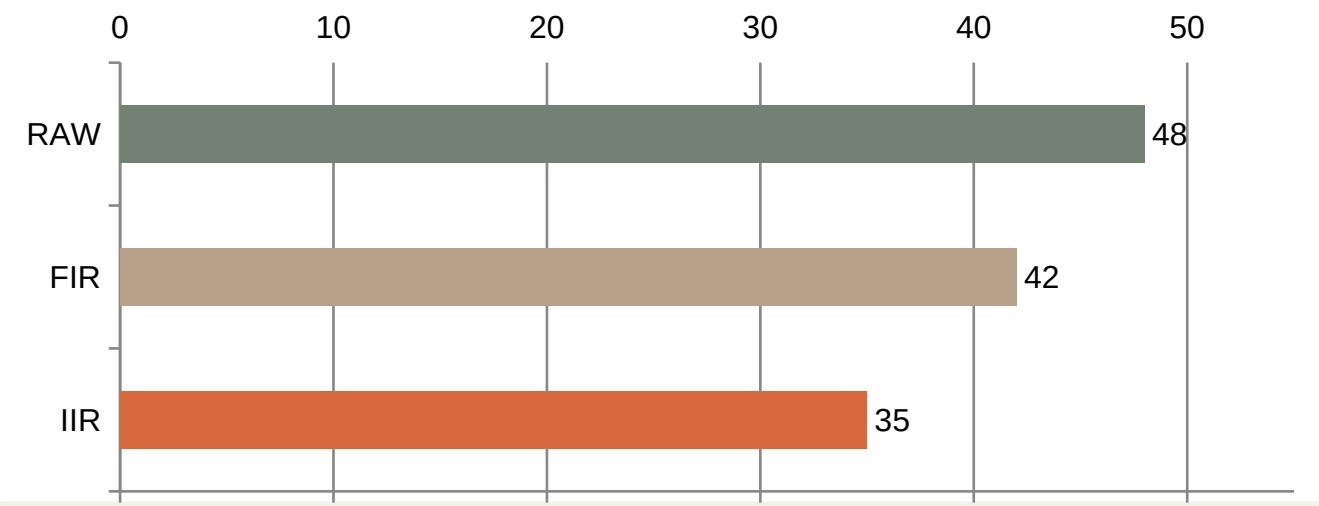
IIR does not lead in every fold.
The mean difference has not been tested for statistical significance.

Source: five-fold evaluation outputs and original notebook chart.

IIR reduces missed severe cases from 48 to 35

Among 480 severe cycles, every missed case was classified as mild rather than normal.

Severe cases not recognized as severe



Method	Missed	Detected	Severe recall
RAW	48	432 / 480	90.00%
FIR	42	438 / 480	91.25%
IIR	35	445 / 480	92.71%

−13

fewer severe cases missed with IIR than RAW

27.1%

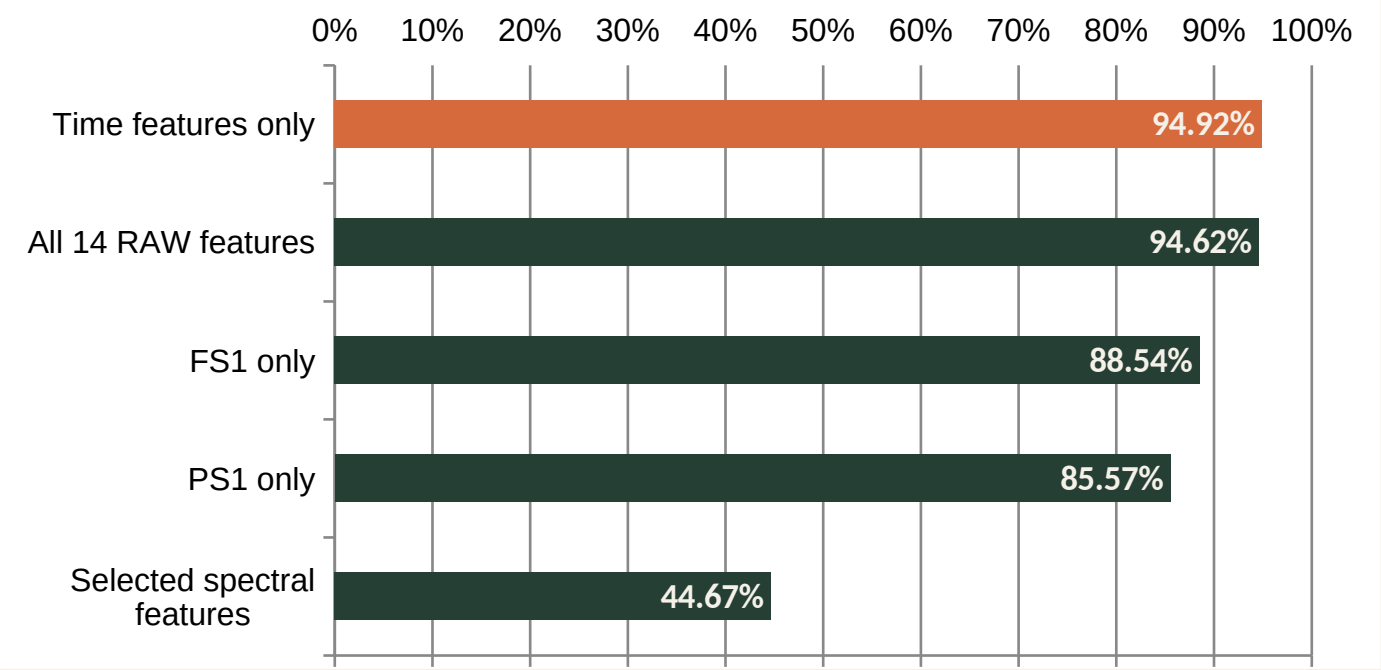
relative reduction in misses
(48 − 35) / 48

Severe precision: 94.74% → 94.68%.
Incorrect severe predictions: 24 → 25.
Severe recall improves, but precision does not.
All counts are held-out predictions.

Source: out-of-fold error counts and class-metrics notebooks.

Time-only features reach 94.92% in the RAW test

Ablation uses unfiltered data and reports mean Macro F1 across five grouped folds.



What the comparison shows

Both sensors help.
PS1 and FS1 together outperform either sensor on its own.

Time features work well.
94.92% time-only vs. 94.62% with all 14 RAW features.

More features do not always help.
The observed gap needs further validation.

The two selected spectral features perform poorly alone.
This does not rule out other frequency-domain approaches.

Source: RAW feature-ablation notebook and grouped validation outputs.

The results are promising, but come from one test rig

Separate measured results from possible explanations and follow-up tests.

Status	Evidence	What remains open
Measured	IIR: mean F1 95.45%; 35 of 480 severe cases missed	Best mean under this test, not in every fold
Hypothesis	Filter attenuation may shape more useful features	Noise was not isolated; mechanism not causal proof
Limitations	One rig; five grouped folds; 0.3, 0.5, 1.0 s explored	Choosing filter settings on these folds may inflate scores
Next experiments	Tune inside training folds; set aside final evaluation	Test other rigs and conditions; measure real-time behavior

No deployment, sensor-noise isolation, or cross-machine validation was performed.

Source: evaluation notebooks and documented study constraints.

What engineers and operations teams need to know

The numbers stay the same. The follow-up questions differ by audience.

Topic	Maintenance / Reliability Engineer	Operations Manager
Sensor data	Check quality, sampling rates, and PS1/FS1 cycle IDs.	Reliable diagnosis depends on reliable sensor data.
DSP	Inspect transients, H(z), and FIR/IIR stability.	Filtering is tested, not assumed to help.
Model results	IIR: mean F1 95.45%; SD 3.23 pp. Keep fold means and OOF apart.	Highest average in this test; field performance is unknown.
Severe cases	35 missed vs. 48 with RAW; all misclassified as mild.	13 fewer severe cases missed in the evaluation.
Trade-offs and limits	Severe precision: 94.74% → 94.68%. One rig; no real-time test.	Better recall, but some false severe predictions remain.
Next tests	Tune on train data; external tests and error reviews.	Run a controlled pilot with engineers before use.

Cost-sensitive analysis would compare error consequences; no financial costs were measured.

Source: held-out predictions, experiment notebooks, and suggested follow-up tests.

IIR helped in this evaluation; field performance is unknown

Data integrity

Check readings, sampling rates, and cycle IDs before building the feature table.

Filter behavior

FIR and IIR reshape transients. H(z) explains the filters, not the hydraulic machine.

Measured performance

IIR: 95.45% mean Macro F1 and 13 fewer severe misses than RAW in held-out testing.

Next steps: separate model selection from final evaluation, then test additional machines before considering deployment.

DATA, METHODS, AND REFERENCES

[1] UCI Hydraulic Systems. DOI 10.24432/C5CW21.
[2] Helwig et al. (2015). I2MTC. DOI 10.1109/I2MTC.2015.7151267.
[3] SciPy. Signal processing and FFT documentation.
[4] scikit-learn. GroupKFold and Random Forest.

Dataset: UCI Hydraulic Systems [1].
Methods: RAW, FIR, IIR, Random Forest [3,4].
Evaluation: five GroupKFold splits.
Outputs: fold metrics, OOF errors, and ablation.
Scope: one laboratory test rig.

Results describe the selected stable cycles and the evaluation setup used in this study.